

Project 3: Drought, Agriculture, and Financial Valuation

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Context

Agricultural activities are highly exposed to drought risk through water availability constraints, yield variability, and input cost pressures. Increasing water stress driven by climate change can affect production volumes, operating costs, and long-term asset productivity. For institutional investors holding equity stakes in agricultural firms, these effects translate into cash flow volatility and valuation risk. The objective of this project is to construct a transparent modelling chain linking drought-related hazards to agronomic impacts and, ultimately, to financial valuation metrics. The analysis is scenario-based and relies primarily on open-access data.

1 Data

We consider a stylised farm of 20 hectares located in the Madrid region (approximately latitude 40.5 and longitude -3.75).

We assume the following production mix:

- 20% sunflower
- 30% maize
- 20% grapes
- 30% wheat

We collect data from several sources:

- SSP climate scenarios and precipitation projections from the Climate Data Store (European Union Copernicus Programme): **SSP585** and **SSP126**, using the Institut Pierre-Simon Laplace (IPSL) and Max Planck Institute (MPI) for Meteorology models, from 2015 to 2100.
- **SPI (Standardized Precipitation Index)** and **SPEI (Standardized Precipitation Evapotranspiration Index)** from the Climate Data Store. We use a 3-month cumulative SPI (resp. SPEI) from 1980 onward, and we average the values for May, June, and July each year to obtain an annual indicator.
- **Crop yields** and production data from FAOSTAT, from 1980 onward.
- **Crop price** data from FAOSTAT, from 1991 onward.
- The **IPAMPA index** (Agricultural Means of Production Purchasing Price Index) from INSEE, used as an input cost index.
- A proxy cost structure for 2020, based on fertilizer and pesticide use from FAOSTAT and prices from the European Commission.

2 Definition of a Stylised Agricultural Firm

Using the FAOSTAT dataset, crop yields are expressed in kg/ha , so we convert them into total tonnes produced on the 20 hectares of the farm.

Using the crop price dataset, we then derive total revenue:

$$Revenue_t = \sum_i w_i \times Yield_{i,t}^{\text{total}} \times Price_{i,t}$$

For the sake of clarity, we are only considering a few cost sources.

The IPAMPA index is expressed with base 100 in 2020. We therefore define a cost structure for 2020 and assume that it evolves proportionally with the IPAMPA index.

- Fertilizers (Nitrogen, P, K): 200 €
- Plant protection products: 65 €
- Seeds: 70 €
- Fuel: 65 €
- Drying / Miscellaneous: 45 €
- Labour: 125 €

This gives a total cost of 570 € per hectare in 2020. Assuming an exchange rate of 1 € = 0.88 \$, this corresponds to:

$$570 \times 0.88 = 501.6 \$$$

per hectare in 2020. We then deduce the historical evolution of costs for the 20-hectare farm using the IPAMPA index:

$$Cost_t = \frac{501.6 \times 20 \times IPAMPA_t}{100}$$

Finally, total margin for each year t is defined as:

$$Margin_t = Revenue_t - Cost_t$$

expressed in US dollars.

Year	Revenue (\$)	Cost (\$)	IPAMPA	Margin (\$)
2005	25572.66	8903.40	78.1	16669.26
2006	30007.06	9154.20	80.3	20852.86
2007	41420.15	9667.20	84.8	31752.95
2008	41620.55	10966.80	96.2	30653.75
2009	33638.76	10271.40	90.1	23367.36
2020	47411.78	11400.00	100.0	36011.78
2021	59135.50	12448.80	109.2	46686.70
2022	53223.25	15241.80	133.7	37981.45
2023	45856.75	14956.80	131.2	30899.95
2024	49040.16	14295.60	125.4	34744.56

Table 1: Farm revenue, costs, and margins (first and last observations)

3 Building a Drought Hazard Indicator

We select the Standardized Precipitation Index (SPI) as a drought stress metric. To define wet and dry thresholds, we first compute the distribution of the SPI (cf. Figure 1).

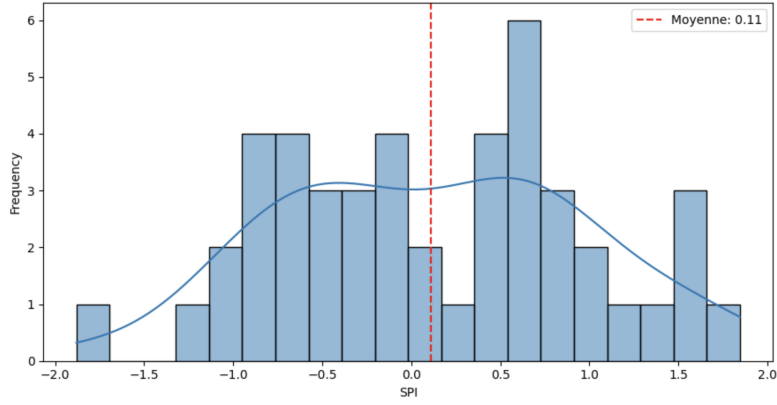


Figure 1: Distribution of the SPI (annual average for May, June, and July)

We choose -0.5 as the threshold in order to remain consistent with the study by Wang, Qianfeng, et al. “Temporal-spatial characteristics of severe drought events and their impact on agriculture on a global scale.” *Quaternary International* 349 (2014): 10–21. Additionally, from the Figure 1, the first quartile of the SPI distribution is -0.59 and the third quartile is 0.66 , which supports the relevance of selecting thresholds of -0.5 and 0.5 .

We then compute a bar plot of the SPI index to highlight drought episodes.

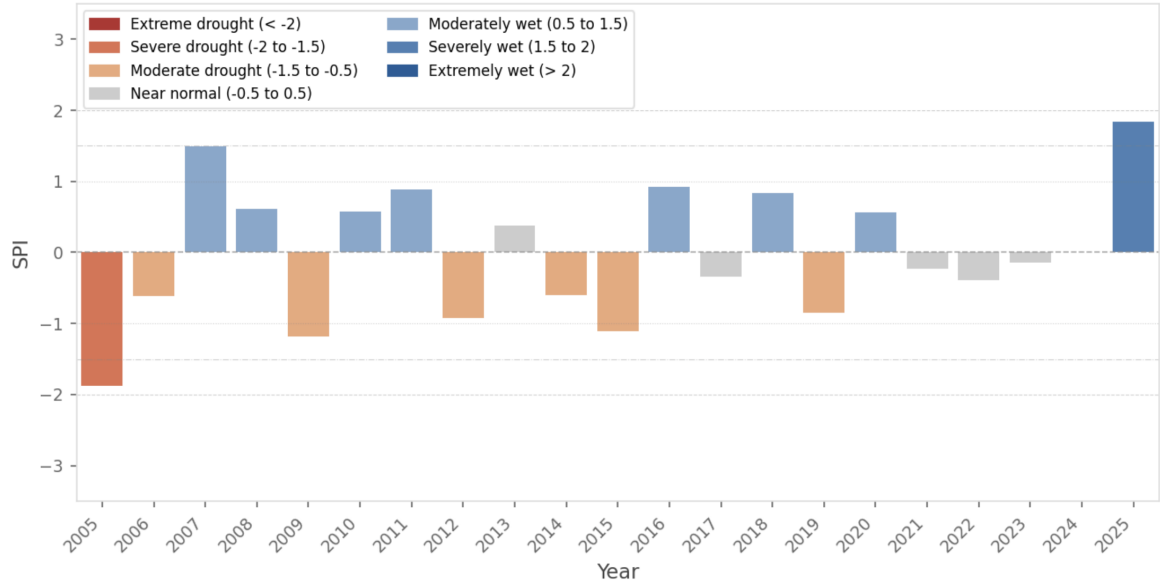


Figure 2: Histogram of the SPI (annual average of SPI-3 for May, June, and July)

The histogram in Figure 2 shows that the annual mean of SPI3 does not show a clear increasing trend in drought over time. Instead, it highlights strong year-to-year variability, with alternating dry and wet years. Some years stand out as particularly dry, such as 2005 and the period from 2012 to 2015, which appears to be the longest dry phase. However, these periods are interrupted by wetter years like 2007, 2008, 2010, 2011, 2016, 2018, 2020, and especially 2025.

Overall, the series suggests recurring drought episodes rather than a steady worsening trend. Since this indicator is based on the average SPI3 for May, June, and July, it smooths short-term extremes.

4 Specifying Agronomic Vulnerability Relationships

We use a graph to illustrate the relationship between drought intensity and yield evolution.

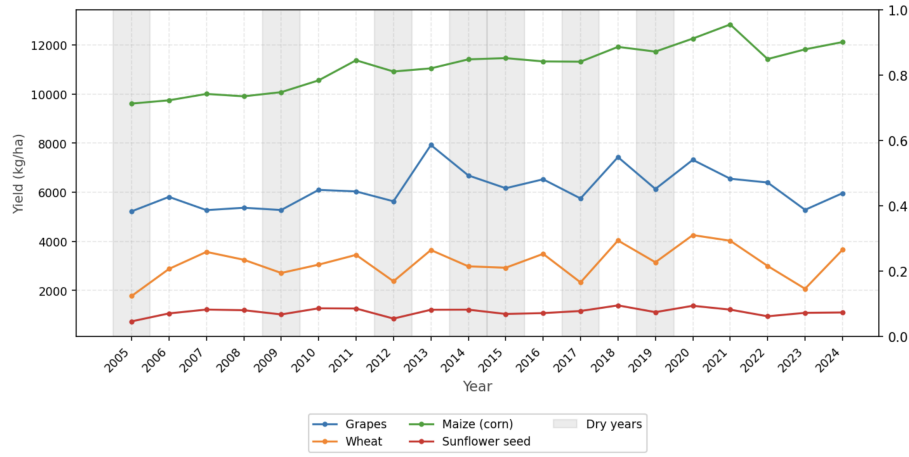


Figure 3: Annual yield by crop (2005–2025)

We observe in Figure 3 that higher drought intensity is associated with significant yield losses, particularly for wheat and grapes (e.g., in 2012 and 2017).

We also model conditional probability density functions of crop yields under wet ($\text{SPI} \geq 0.5$; blue) and dry ($\text{SPI} < -0.5$; orange) conditions for each crop on the farm.

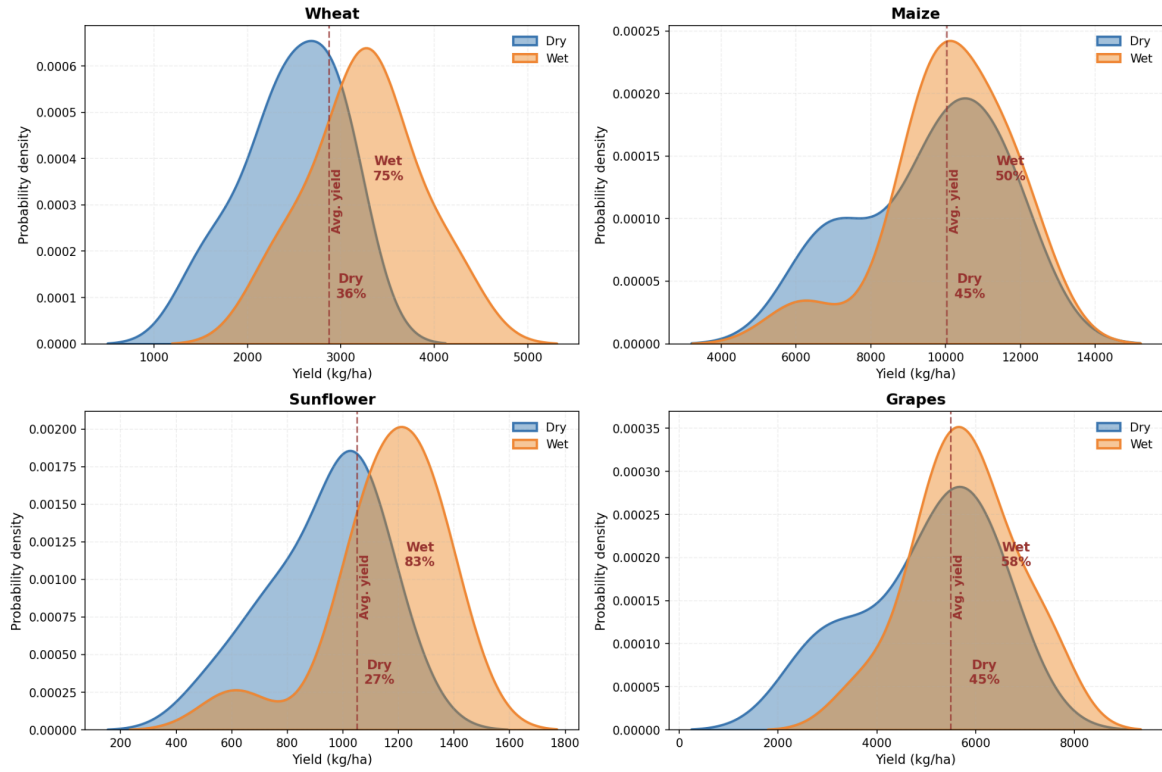


Figure 4: Conditional density of crop yields under dry vs. wet conditions (SPI categories), 2005–2025

Figure 4 shows that for each crop since 1980, the probability of yields exceeding the annual average is higher in wet conditions than in dry conditions. For example, the probability that annual wheat production exceeds its average yield is 27% in dry conditions ($\text{SPI} < -0.5$) and 83% in wet conditions ($\text{SPI} \geq 0.5$).

Overall, drought risk for annual production appears to be lower for maize and grapes than for wheat and sunflower.

5 Translating Agronomic Impacts into Financial Impacts

5.1 Computation of projected SPI

We now study the impact of drought intensity on financial flows by computing the variation (%) in revenue, costs, and margins.

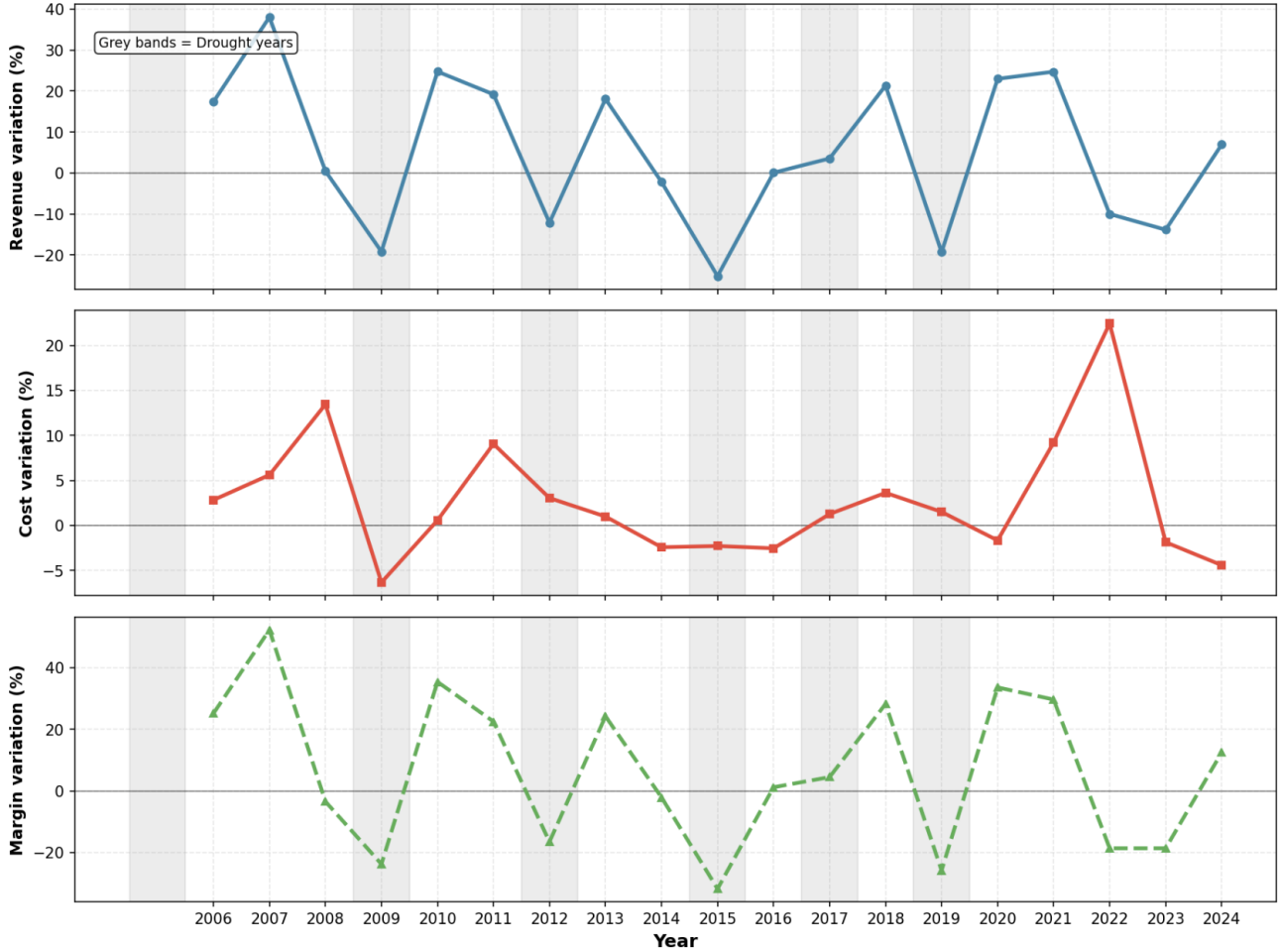


Figure 5: Farm financial variation (2005–2025): costs and margins during drought years

We clearly observe in Figure 5 that drought episodes negatively affect revenue variation, as well as total margin variation. This is consistent with the adverse impact of drought conditions on crop yields. Regarding cost variation, the effect is more mixed. In some drought years (e.g., 2009), costs decrease, but this pattern is not systematic (e.g., 2015, 2017, and 2019). When production declines, certain variable costs, such as labor and pesticide use, may also decrease. However, input prices, particularly energy, can increase, leading to higher overall costs even during drought periods (e.g., 2017). Additionally, external factors such as the Ukraine war and inflation around 2022 contributed to rising costs.

6 Conduct Scenario Analysis

We apply two climate scenarios, the best (SSP126) and the worst (SSP585), to modify the drought index (SPI). For both scenarios, we select two different models: the French Institut Pierre Simon Laplace (IPSL) and the German Max Planck Institute (MPI).

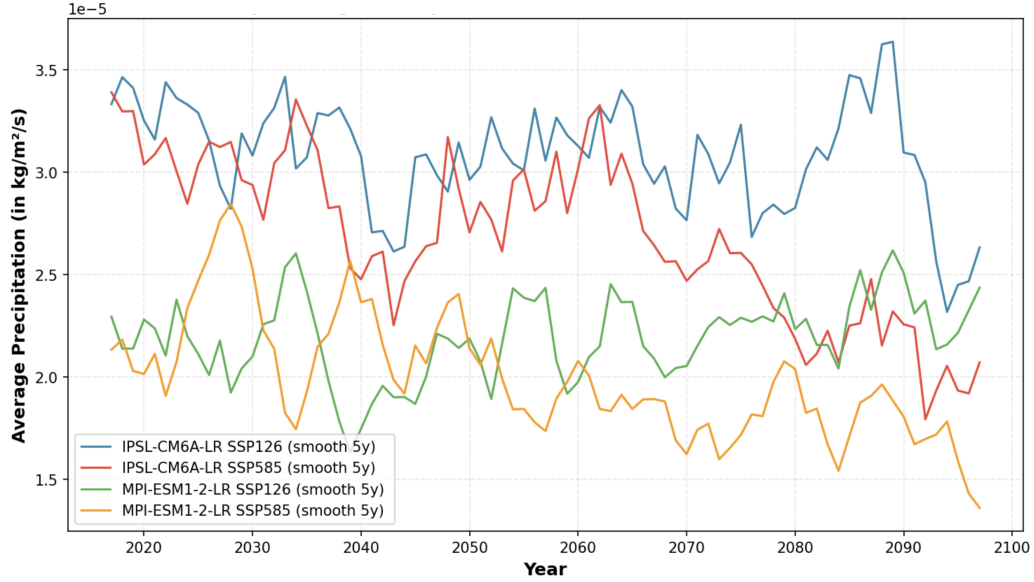


Figure 6: Monthly average precipitation under different climate scenarios

We can clearly observe in Figure 6 that, in both models, the SSP126 scenario generally implies higher precipitation levels than the SSP585 scenario, particularly from around 2050 onwards. This is consistent, as SSP585 represents a more severe climate change pathway with stronger warming and associated drying effects in some regions.

However, there is a noticeable difference in magnitude between the two models: the IPSL model shows higher overall precipitation levels compared to the MPI model.

Within the MPI model, we can observe some inconsistencies between the two scenarios. For instance, around 2035, precipitation decreases under SSP585 while it increases under SSP126, leading to a divergence that is not entirely smooth. Given these discrepancies, we will focus the analysis on the **IPSL model**, which appears more consistent for our purposes.

We compute the SPI3 from the precipitation predictions over 2015-2100. For each scenarios we compute monthly SPI-3 values over the period 2015–2100. Then, we focus on the May–July period, which corresponds to the main crop growing season, and compute the annual SPI as the average of these three months.

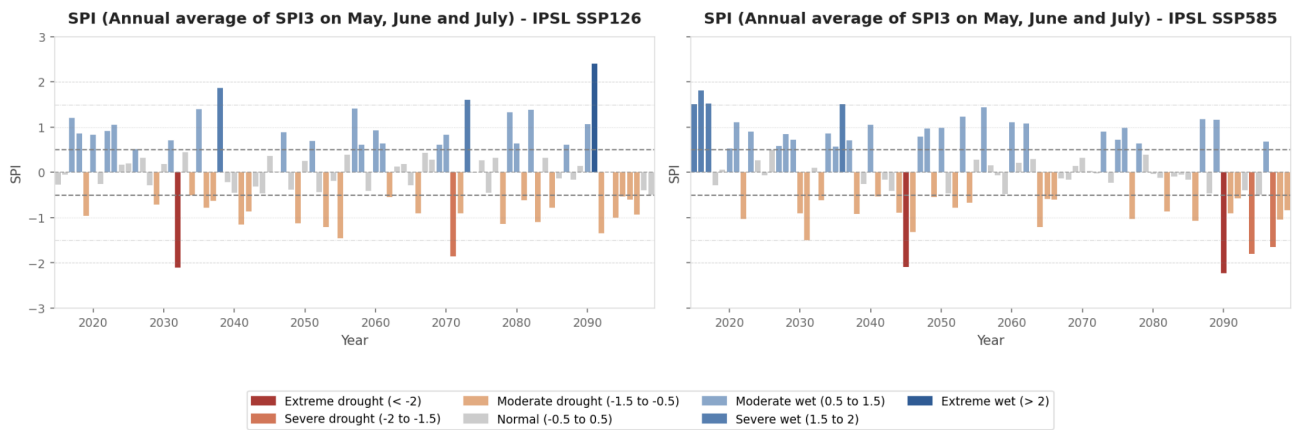


Figure 7: Annual Average SPI3 under IPSL Climate Scenarios

We can see in Figure 7 that the histogram on the left corresponds to the SSP126 scenario, while the one on the right corresponds to the SSP585 scenario. We note that the SPI values are computed separately within each scenario, using each scenario's own historical period from 2015 onward as the reference distribution.

We can clearly observe in both scenarios that the frequency of drought episodes increases significantly after 2030 compared with the 2015–2030 period. We also note that under the more severe SSP585 scenario, extreme wet episodes appear to be less frequent than under SSP126, particularly from around 2070 onward.

6.1 Modeling of the Projected Revenues

At this point, we compute the financial outcomes under each scenario. We assume a linear relationship between SPI and farm revenue. We therefore estimate a linear regression model in which revenue is the dependent variable and the explanatory variable is the SPI.

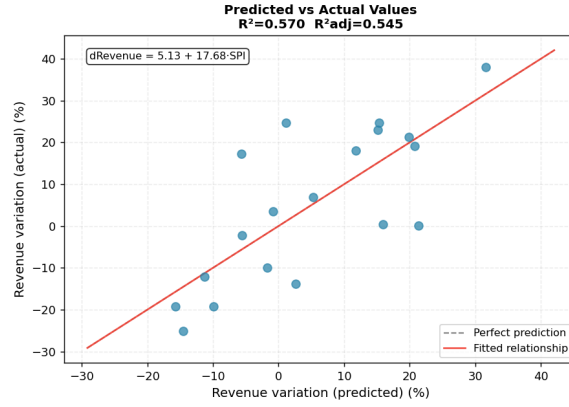


Figure 8: Linear regression on revenue variation: predicted vs. actual values

The number of observations is relatively small; however, the graphical analysis of the Figure 8 suggests a strong linear relationship between SPI levels and farm revenue variations. Therefore, for simplicity and interpretability, we adopt a linear regression model.

7 Perform valuation analysis

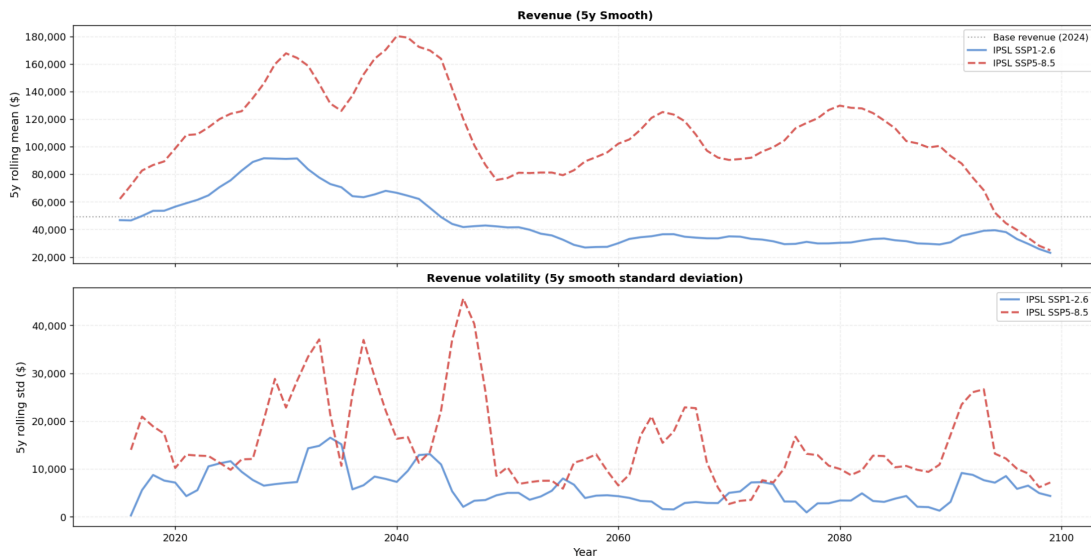


Figure 9: Predicted revenues from the regression model

The revenue projections represented in the Figure 9 suggest that climate change affects not only the level of farm income, but also its stability over time. In the upper graph, we present the projected

revenues, while the lower graph shows the volatility of these revenues (5-year smoothing). Under the moderate scenario (IPSL SSP1-2.6), projected revenues initially increase and then gradually decline, while remaining relatively stable in terms of volatility.

In contrast, the more adverse scenario (IPSL SSP5-8.5) leads to higher average revenues over much of the period, but also to much larger year-to-year fluctuations, as shown by the rolling standard deviation.

The high revenues around 2040 can be explained by the SPI projections under the SSP5-8.5 scenario. Climate change does not only lead to droughts, but also to periods of increased rainfall between 2025 and 2045, which may be favourable for crop growth.

From an economic perspective, this is not favourable for the farmer. Highly volatile revenues make planning more difficult, reduce cash-flow visibility, complicate labour and input management, and may require frequent adjustments. In practice, a farmer is unlikely to prefer an unstable stream of high revenues over a more stable and predictable income.

These results suggest that focusing only on average revenues can be misleading. While the severe scenario may appear attractive in terms of average income, it actually implies a more uncertain and risky economic environment for the farm.

8 Sensitivity and Uncertainty Analysis

8.1 Test alternative drought indicator

We now test alternative drought indicators. We study the Standardized Precipitation Evapotranspiration Index (SPEI). This index integrates the effect of temperature and evaporation, and can capture droughts related to high temperatures.

We use the same thresholds for the SPEI index as for the SPI index: -0.5 and 0.5 for dry and wet conditions, respectively.

We are interested in comparing the risk associated with crops under different drought indices and in assessing how agricultural decisions may differ depending on the index used.

We compare the conditional probability density functions of sunflower yields in wet (blue) and dry (orange) conditions using both SPEI (left) and SPI (right).

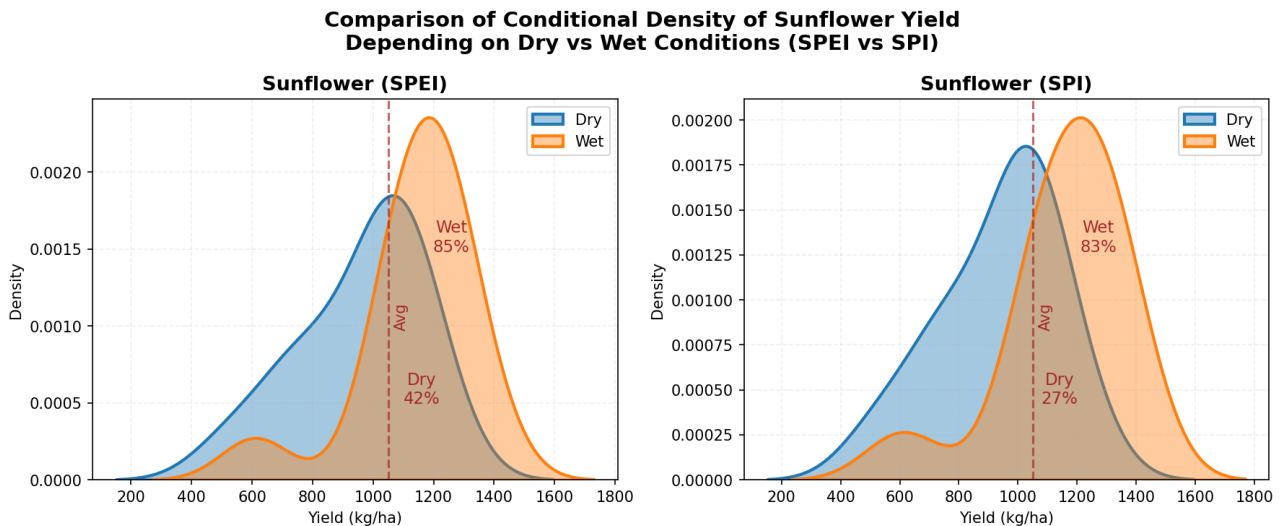


Figure 10: Comparison of conditional density of sunflower yield

The Figure 10 shows the probability that annual sunflower production exceeds its average yield is 42% in dry conditions when using SPEI, whereas it is only 27% when using SPI. The drought risk for sunflower production appears lower when using SPEI than when using SPI.

We can therefore deduce that sunflower may be considered less risky when using SPEI rather than SPI. As a result, a farmer might allocate a larger share of land to sunflower if decisions are based on SPEI instead of SPI.

8.2 Crop switching

We first regress the historical SPI index on the yields of the different crops in order to measure how crop productivity reacts to drought conditions.

The regression results show that maize and grapes have relatively low explanatory power (cf. Figure 2), with low R^2 values, suggesting that their yields are less strongly driven by SPI fluctuations.

	Crop	beta_SPI	p_value	R^2	n_obs	mean_yield
0	Maize	+165.56	0.502	0.025	20	11,142
1	Grapes	+279.36	0.173	0.100	20	6,140
2	Wheat	+513.14	0.001	0.451	20	3,129
3	Sunflower	+137.53	0.000	0.551	20	1,128

Table 2: Regression results by crop

This is consistent with the idea that these crops are more robust to drought conditions than the others in the sample. We therefore select maize and grapes as the crops to prioritize in dry years and model an adaptive crop-switching strategy toward these two productions.

We now show the efficiency of land reallocation across crops, then we compare the farmer's historical revenue over 2005–2022 with the revenue that would have been obtained if this reallocation strategy had been used.

Methodology:

- We use a dummy variable equal to 1 in drought years ($\text{SPI} < -0.5$) and 0 otherwise.
- In drought years, we change the crop allocation and give a larger share of land to grapes and maize, which are considered more suitable for summer growing conditions. Otherwise, we keep the normal allocation: Grapes 20%, Maize (corn) 30%, Sunflower seed 20%, and Wheat 30%.

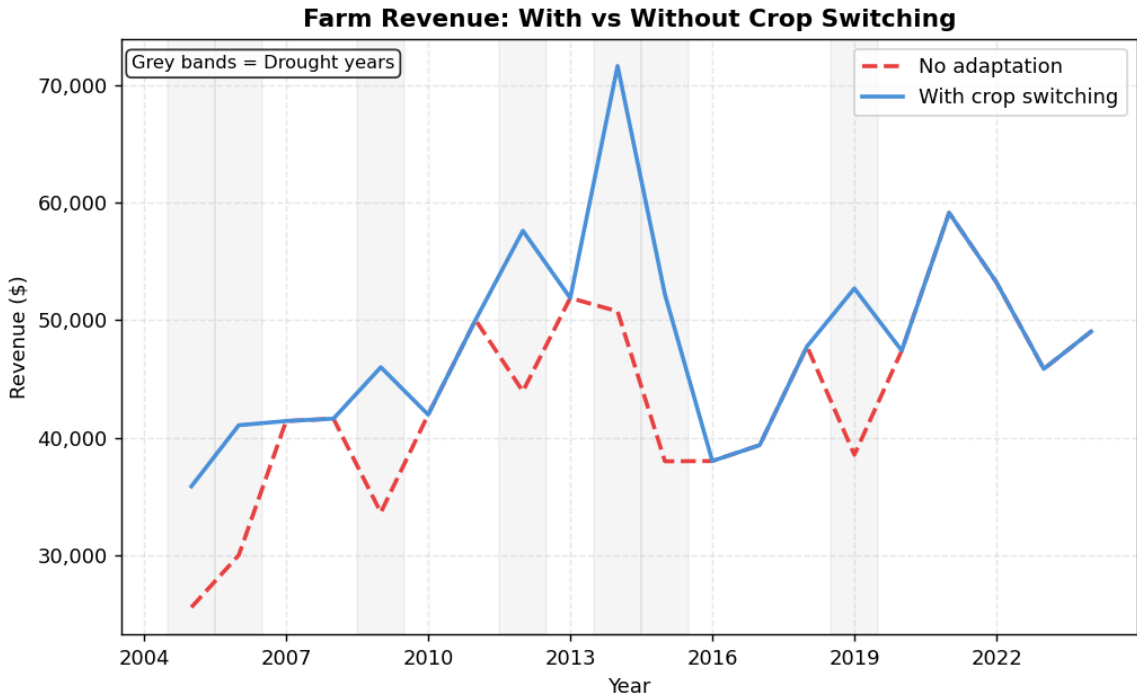


Figure 11: Adaptive crop switching based on historical crop sensitivity to SPI

The figure 11 compares the farmer’s historical revenue path with and without an adaptive crop-switching strategy. In dry years, we reallocate production toward crops assumed to be more resilient to drought, using the following weights: 35% for grapes, 35% for maize, 15% for sunflower seed, and 15% for wheat.

Overall, the graph suggests that crop switching could have acted as an effective risk-mitigation strategy, allowing the farmer to preserve higher revenues and, in several years, generate a clear income gain relative to the observed non-adaptive case. For example, in 2019, with adaptation, the farmer could have increased total crop revenue by about \$12,000.

9 Conclusion

Farm revenues are likely to be strongly affected by climate change, especially due to drought risk, regardless of the scenario considered. The main issue is the increasing instability of farmers’ income, particularly under a severe scenario (SSP5-8.5), which is close to the current climate trajectory.

To better understand these risks, farmers can use different drought indicators such as SPI or SPEI, as well as prediction models. The more accurate these models are, the better farmers can anticipate potential losses and manage risk. The linear regression we used can be improved, for example, by using copulas.

In addition, farmers can reduce their exposure to risk by choosing crops that are more resilient to drought.

10 References

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